Complex Numbers

Chapter Overview

Solution 2

1. Understand and manipulate complex numbers

Write $(4+i)/(3-i)$ in the form $a+bi$.

2. Solve quadratics over the complex numbers

Solve $x^2+3x+5=0$.

3. Solve cubic and quartic equations

Given $-2+i$ is a root of $x^3+3x^2+x-5=0$, determine the other roots.

Why complex numbers? Solution 3

The quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

has no real value when the discriminant is negative. Introducing i, with $i^2 = -1$, lets us express these roots.

Complex numbers arose in sixteenth-century work on polynomial equations, including Cardano's work.



$i = \sqrt{-1}$.

An imaginary number has form bi, $b \in \mathbb{R}$, such as $i, 3i, -2i, \pi i$.

A complex number has form $a + bi$, $a, b \in \mathbb{R}$, such as $1 + i, 3 - 2i$. Its real part is a and its imaginary part is b.

Notation

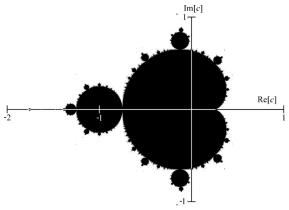
Coefficients usually precede constants and variables, as in $2\pi x$ or $3\pi i x$. Both $a + bi$ and $a + ib$ are valid: multiplication commutes.

The conventional form of Euler's formula is $e^{i\theta} = \cos\theta + i\sin\theta$, with special case $e^{i\pi} + 1 = 0$. Writing $e^{\theta i}$ is mathematically equivalent.

Other Practical Applications Solution 4

1. Fractals

The Mandelbrot set is a famous fractal. We will investigate its recurrence later in this chapter.



2. Analytic number theory

Number theory studies integers. Analytic methods use real and complex variables; for example, the Riemann zeta function accepts complex inputs and is linked to the distribution of primes.

3. Physics and engineering

Applications include signal analysis, quantum mechanics, fluid dynamics, relativity and control theory.

Complex Number Basics

Solution 5

Write in terms of i:

$\sqrt{-36}$

$\sqrt{-36} = \sqrt{36}\,i = 6i.$

$\sqrt{-4}$

$2i.$

$\sqrt{-7}$

$\sqrt{7}\,i.$

Simplify:

$(2+3i)+(4+i)$

4

$(2+4)+(3+1)i = 6+4i.$

$i-3(2-i)$

$i-6+3i = -6+4i.$

$\frac{10+4i}{2}$

$5+2i.$

Solving Quadratic Equations

Solution 6

Solve $z^2+25=0$.

$z^2 = -25 \Rightarrow z = \pm 5i.$

Solve $z^2+3z+5=0$.

$a=1, b=3, c=5.$

$$z = \frac{-3 \pm \sqrt{9-20}}{2} = -\frac{3}{2} \pm \frac{\sqrt{11}}{2}i.$$

Notation note

We often use x for real variables, n for integers and z for complex variables.

Exercises 1A-1B Pearson Core Pure 1, p. 3, 4-5

Solution 7

4. Write in the form $a+bi$, with a, b simplified surds.

a. $(4-2i)/\sqrt{2}$

b. $(2-6i)/(1+\sqrt{3})$

4a

$$\frac{4-2i}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}-2\sqrt{2}i}{2} = 2\sqrt{2}-\sqrt{2}i.$$

4b

$$\frac{2-6i}{1+\sqrt{3}}\frac{1-\sqrt{3}}{1-\sqrt{3}} = \frac{2-2\sqrt{3}-6i+6\sqrt{3}i}{-2}$$
$$= (-1+\sqrt{3})+(3-3\sqrt{3})i.$$

6. Given $z_1 = a+9i, z_2 = -3+bi$ and $z_2 - z_1 = 7+2i$, find real a, b.

6

$$7+2i = (-3+bi)-(a+9i) = (-3-a)+(b-9)i.$$

Equate real and imaginary parts:

$$7 = -3-a \Rightarrow a = -10, \quad 2 = b-9 \Rightarrow b = 11.$$

Exercises 1A Pearson Core Pure 1, p. 3

Solution 8

7. $z_1 = 4+i, z_2 = 7-3i$. Find, in the form $a+bi$: (a) $z_1 - z_2$, (b) $4z_2$, (c) $2z_1 + 5z_2$.

7a

$$(4+i)-(7-3i) = (4-7)+(1+3)i = -3+4i.$$

7b $4(7-3i) = 28-12i.$

7c

$$2(4+i)+5(7-3i) = 8+2i+35-15i = 43-13i.$$

8. Given $z = a+bi, w = a-bi, a, b \in \mathbb{R}$, show (a) $z+w$ is real, (b) $z-w$ is imaginary.

8a

$$(a+bi)+(a-bi) = 2a \in \mathbb{R}.$$

8b

$$(a+bi)-(a-bi) = 2bi.$$

This has real part zero (and may be zero).

Exercises 1B Pearson Core Pure 1, p. 4-5

Solution 9

2. Solve, giving answers in the form $a + bi$.

a. $(z-3)^2 - 9 = -16$

2a

$(z-3)^2 = -7 \Rightarrow z - 3 = \pm\sqrt{7}i \Rightarrow z = 3 \pm \sqrt{7}i.$

b. $2(z-7)^2 + 30 = 6$

2b

$(z-7)^2 = -12 \Rightarrow z - 7 = \pm 2\sqrt{3}i \Rightarrow z = 7 \pm 2\sqrt{3}i.$

c. $16(z+1)^2 + 11 = 2$

2c

$(z+1)^2 = -9/16 \Rightarrow z + 1 = \pm 3i/4.$

Hence $z = -1 \pm 3i/4.$

6. $z^2 + bz + 11 = 0$, $b \in \mathbb{R}$, has distinct non-real roots. Find the possible values of b.

6

Negative discriminant:

$b^2 - 44 < 0 \Rightarrow b^2 < 44 \Rightarrow -2\sqrt{11} < b < 2\sqrt{11}.$

Multiplying Complex Numbers

Solution 10

$i^2 = -1$. Expand into $a + bi$:

1. $(2 + 3i)(3 - 2i)$

1 $6 - 4i + 9i - 6i^2 = 6 + 5i + 6 = 12 + 5i.$

2. $(5 - 3i)^2$

2

$25 - 30i + 9i^2 = 25 - 30i - 9 = 16 - 30i.$

Find $i^3, i^4, i^{101}, (3i)^5$.

$i^3 = i^2 i = -i.$

$i^4 = i^2 i^2 = 1.$

The powers repeat every four: $i, -1, -i, 1$. For i^k, use the remainder when k is divided by 4.

6. Power 101

$i^{101} = (i^4)^{25} i = i.$

7. Scaled power

$(3i)^5 = 3^5 i^5 = 243i.$

Test Your Understanding

Solution 11

Edexcel FP1 June. Given $z = 2 - 3i$, show $z^2 = -5 - 12i$.

$(2 - 3i)^2 = 4 - 12i + 9i^2 = 4 - 12i - 9 = -5 - 12i.$

Edexcel FP1 June 2010. Expand and simplify $(1 + i)^3$.

$1 + 3i + 3i^2 + i^3 = 1 + 3i - 3 - i = -2 + 2i.$

Exercise 1C Pearson Core Pure 1, p. 6

Solution 12

3. Given $(a + 3i)(1 + bi) = 25 - 39i$, find two possible real pairs (a, b).

1. Equate

$a + abi + 3i + 3bi^2 = (a - 3b) + (ab + 3)i.$

Equate parts:

$a - 3b = 25 \Rightarrow a = 3b + 25, \qquad ab + 3 = -39.$

2. Solve

$(3b + 25)b + 3 = -39 \Rightarrow 3b^2 + 25b + 42 = 0$

$(3b + 7)(b + 6) = 0 \Rightarrow b = -7/3 \text{ or } -6.$

3. Pairs

If $b = -6$, $-6a + 3 = -39$, so $a = 7$.

If $b = -7/3$, $-7a/3 + 3 = -39$, so $a = 18$.

The pairs are $(7, -6)$ and $(18, -7/3)$.

Exercise 1C Pearson Core Pure 1, p. 6

Solution 13

5. Express $(1+i)^6$ in the form $a-bi$, where a, b are integers.

Use the binomial expansion:

$(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3$

$+15a^2b^4 + 6ab^5 + b^6$.

5

With $a=1, b=i$:

$(1+i)^6 = 1 + 6i + 15i^2 + 20i^3 + 15i^4 + 6i^5 + i^6$.

Use $i^2=-1, i^3=-i, i^4=1, i^5=i, i^6=-1$:

$= 1 + 6i - 15 - 20i + 15 + 6i - 1 = -8i$.

In the requested form $a-bi$, $a=0, b=8$.

Exercise 1C Pearson Core Pure 1, p. 6

Solution 14

7. $f(z) = 2z^2 - z + 8$. Find (a) $f(2i)$, (b) $f(3-6i)$.

7a $2(2i)^2 - 2i + 8 = 8i^2 - 2i + 8 = -2i$.

7b

$2(3-6i)^2 - (3-6i) + 8$

$= 2(9 - 36i + 36i^2) - 3 + 6i + 8$

$= 18 - 72i - 72 - 3 + 6i + 8 = -49 - 66i$.

8. For $f(z) = z^2 - 2z + 17$, show $z = 1 - 4i$ solves $f(z) = 0$.

8

$f(1-4i) = (1-4i)^2 - 2(1-4i) + 17$

$= 1 - 8i + 16i^2 - 2 + 8i + 17$

$= 1 - 16 - 2 + 17 = 0$.

For Interest: The Mandelbrot Set

Solution 15

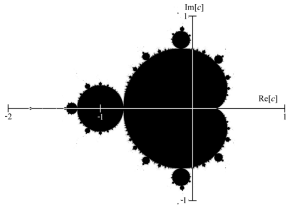
How does the Mandelbrot set work?

Recurrences: $x_{n+1} = x_n + 2, x_1 = 0$ gives $0, 2, 4, \ldots$, diverging.

$x_{n+1} = 5 + x_n/2, x_1 = 0$ gives $0, 5, 7.5, 8.75, \ldots$, converging to 10.

For the Mandelbrot set, use

$$z_{n+1} = z_n^2 + c, \quad z_1 = 0.$$



For $c = 1$:

$z_2 = 0^2 + 1 = 1,$

$z_3 = 1^2 + 1 = 2,$

$z_4 = 2^2 + 1 = 5.$

The sequence is unbounded.

For $c = i$:

$z_2 = 0^2 + i = i,$

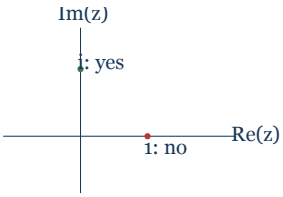
$z_3 = i^2 + i = -1 + i,$

$z_4 = (-1 + i)^2 + i$

$= 1 - 2i - 1 + i = -i,$

$z_5 = (-i)^2 + i = -1 + i.$

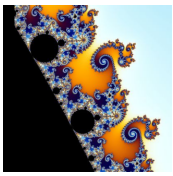
It then alternates and is bounded.



Plot $a + bi$ at (a, b) on an Argand diagram (chapter 2). The Mandelbrot set consists of the values of c whose sequence stays bounded. Bounded sequences need not converge or repeat.

Thus i, plotted at $(0, 1)$, belongs to the set; 1, at $(1, 0)$, does not.

Considering every c produces the fractal boundary. Colours outside the set can show escape speed; black indicates bounded behaviour in this illustration.



Complex Conjugation

Solution 16

Let $x = 3 + \sqrt{2}, x^* = 3 - \sqrt{2}$. Here the star labels the algebraic surd conjugate, not the complex conjugate: both numbers are real. Find their sum and product.

$x + x^* = 6.$

$xx^* = 9 - 2 = 7.$

Both are rational and free of surds. Their product is useful when rationalising a denominator.

Let $z = 3 + 2i, z^* = 3 - 2i$. Find their sum and product.

4. Complex sum $z + z^* = 6.$

5. Complex product

$zz^* = (3 + 2i)(3 - 2i) = 9 - (2i)^2 = 9 + 4 = 13.$

Both results are real. This gives a way to make a denominator real, allowing division of complex numbers.

Complex Conjugation

Solution 17

If $z = a + bi$, its complex conjugate is $z^* = a - bi$.

Write $(5 + 4i)/(2 - 3i)$ in the form $a + bi$.

2. Division

$$\frac{5 + 4i}{2 - 3i}\frac{2 + 3i}{2 + 3i} = \frac{10 + 15i + 8i - 12}{4 - (3i)^2}$$
$$= \frac{-2 + 23i}{13} = -\frac{2}{13} + \frac{23}{13}i.$$

Multiply numerator and denominator by the conjugate of the denominator, just as for surds.

Remember:

$(a + bi)(a - bi) = a^2 + b^2$.

Calculator tips from the source

On the calculator shown in the original course, choose Mode → Complex, use ENG for i, and OPTN for conjugate, real-part and imaginary-part functions. Menu names vary by model. Complex mode can check products, quotients and powers.

Test Your Understanding

Solution 18

FP1 January 2009 Q9(a).

Given $z_1 = 3 + 2i$, $z_2 = (12 - 5i)/z_1$, find z_2 in the form $a + ib$, $a, b \in \mathbb{R}$.

9(a)

$$z_2 = \frac{12 - 5i}{3 + 2i}\frac{3 - 2i}{3 - 2i} = \frac{36 - 24i - 15i - 10}{9 + 4}$$
$$= \frac{26 - 39i}{13} = 2 - 3i.$$

Exercise 1D Pearson Core Pure 1, p. 7-8 — Solution 19

2. Find $z + z^*$ and zz^* for each value.

a. $z = 6 - 3i$

2a

$z + z^* = 12;$

$zz^* = (6 - 3i)(6 + 3i) = 36 + 18i - 18i - 9i^2 = 45.$

b. $z = 10 + 5i$

2b

$z + z^* = 20;$

$zz^* = (10 + 5i)(10 - 5i) = 100 - 25i^2 = 125.$

c. $z = 3/4 + i/4$

2c

$z + z^* = 3/2;$

$zz^* = (3/4)^2 - (i/4)^2 = 9/16 + 1/16 = 5/8.$

d. $z = \sqrt{5} - 3\sqrt{5}i$

2d

$z + z^* = 2\sqrt{5};$

$zz^* = (\sqrt{5})^2 - (3\sqrt{5}i)^2 = 5 + 45 = 50.$

4. Write $(3 - 4i)^2/(1 + i)$ in the form $x + iy$.

4

$$\frac{9 - 24i + 16i^2}{1 + i}\frac{1 - i}{1 - i} = \frac{(-7 - 24i)(1 - i)}{2}$$
$$= \frac{-7 + 7i - 24i + 24i^2}{2} = -\frac{31}{2} - \frac{17}{2}i.$$

Exercise 1D Pearson Core Pure 1, p. 7-8 — Solution 20

6. Given $(5 + 2i)/z = 2 - i$, find z in the form $a + bi$.

6

$$z = \frac{5 + 2i}{2 - i}\frac{2 + i}{2 + i} = \frac{10 + 5i + 4i + 2i^2}{4 - i^2}$$
$$= \frac{8 + 9i}{5} = \frac{8}{5} + \frac{9}{5}i.$$

Edexcel FP1 June 2013 Q9.

For $z = x + iy$, $x, y \in \mathbb{R}$, solve $(3 - i)z^* + 2iz = 9 - i$.

Exam

$(3 - i)(x - iy) + 2i(x + iy) = 9 - i$

$3x - 3iy - ix - y + 2ix - 2y = 9 - i$

$(3x - 3y) + (x - 3y)i = 9 - i.$

So $x - y = 3, x - 3y = -1$. Subtract: $2y = 4$; hence $y = 2, x = 5$.

Roots of Quadratics

Solution 21

$x^2 + 4x - 5 = 0$ factors as $(x + 5)(x - 1) = 0$, so its roots are $-5, 1$. Conversely, those roots give the two factors by the factor theorem.

If α, β are the roots, the monic quadratic is $(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta$.

For a polynomial with real coefficients, non-real roots occur in complex conjugate pairs.

If a real-coefficient quadratic has root $3 + 4i$, what is its other root?

It is $3 - 4i$. Can you prove why?

The non-real condition matters: two real roots need not be equal.

Why Non-real Roots Come in Conjugate Pairs

Solution 22

Prove non-real roots of a quadratic with real coefficients occur in conjugate pairs.

Proof 1: quadratic formula

For real a, b, c, $a \neq 0$, suppose $b^2 - 4ac < 0$.

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$
$$= -\frac{b}{2a} \pm \frac{\sqrt{4ac - b^2}}{2a} i.$$

The real parts are equal and imaginary parts opposite: the roots are conjugates.

Proof 2: sum and product

$\alpha + \beta$ and $\alpha\beta$ must be real. Put $\alpha = a_1 + b_1 i, \beta = a_2 + b_2 i$.

$\alpha + \beta = (a_1 + a_2) + (b_1 + b_2)i$. From the sum, $b_1 + b_2 = 0$, so $b_2 = -b_1$.

$\alpha\beta = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$.

Its imaginary part vanishes:

$-a_1 b_1 + a_2 b_1 = b_1(a_2 - a_1) = 0$.

Either $b_1 = b_2 = 0$ (both roots real), or $a_2 = a_1$ and $b_2 = -b_1$, proving conjugacy for non-real roots.

Example Question

Solution 23

Given $\alpha = 7 + 2i$ is a root of a quadratic with real coefficients, (a) state the other root β, (b) find the quadratic equation.

a $\beta = 7 - 2i.$

The sum-and-product approach is shorter; remember $(a + bi)(a - bi) = a^2 + b^2$.

b expansion

Expansion

$(z - (7 - 2i))(z - (7 + 2i)) = 0.$

$z^2 - (7 + 2i)z - (7 - 2i)z + (7 - 2i)(7 + 2i) = 0$

$z^2 - 7z - 2iz - 7z + 2iz + 49 - 14i + 14i + 4 = 0$

$z^2 - 14z + 53 = 0.$

b quick method

Sum and product

$\alpha + \beta = 14, \alpha\beta = 7^2 + 2^2 = 53.$

$z^2 - (\alpha + \beta)z + \alpha\beta = z^2 - 14z + 53 = 0.$

Test Your Understanding

Solution 24

FP1 January 2011 Q4.

$2 - 4i$ is a root of $z^2 + pz + q = 0$, with real p, q.

a. State the other root.

b. Find p, q.

a $2 + 4i.$

b

$(z - (2 - 4i))(z - (2 + 4i)) = (z - 2)^2 + 16$

$= z^2 - 4z + 20.$

Thus $p = -4, q = 20.$

Exercise 1E Pearson Core Pure 1, p. 9-10 — Solution 25

3. Given $2+3i$ is a root of a real-coefficient quadratic, (a) state the other root, (b) find the equation in the form $z^2+bz+c=0$.

3a $2-3i$.

3b

Sum 4, product $2^2+3^2=13$.

$z^2-4z+13=0$.

4. Given $5-i$ is a root of $z^2+pz+q=0$, $p,q\in\mathbb{R}$, (a) state the other root, (b) find p,q.

4a $5+i$.

4b

$(z-(5-i))(z-(5+i))=(z-5)^2+1$

$=z^2-10z+26$.

Thus $p=-10, q=26$.

Exercise 1E Pearson Core Pure 1, p. 9-10 — Solution 26

5. Given $-5+4i$ is a root of $z^2+bz+c=0$, $b,c\in\mathbb{R}$, find b,c.

5

The other root is $-5-4i$.

$(z+5-4i)(z+5+4i)=(z+5)^2+16$

$=z^2+10z+41$.

Hence $b=10, c=41$.

6. Given $1+2i$ is a root of a real-coefficient quadratic, find its equation as $z^2+bz+c=0$ with integer b,c.

6

The other root is $1-2i$.

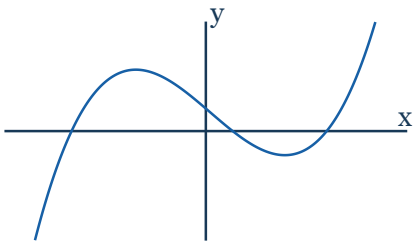
$\alpha+\beta=2, \quad \alpha\beta=(1+2i)(1-2i)=1+4=5$.

Hence $z^2-2z+5=0$.

Roots of Cubic and Quartic Equations

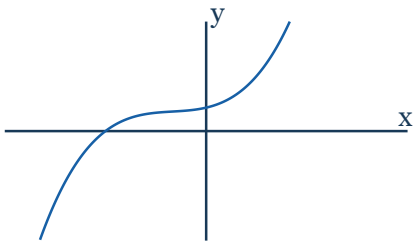
Solution 27

For real-coefficient polynomials, non-real roots occur in conjugate pairs. The fundamental theorem of algebra gives three complex roots counting multiplicity for a cubic.



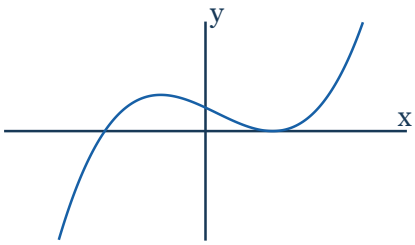
Comment on the roots.

Three real roots.



Comment on the roots.

One real root and two non-real conjugate roots.



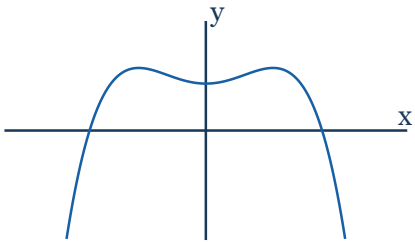
Comment on the roots.

Three real roots counting multiplicity; one is repeated.

Roots of Cubic and Quartic Equations

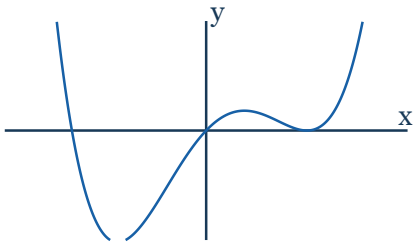
Solution 28

For real-coefficient polynomials, non-real roots occur in conjugate pairs. The fundamental theorem of algebra gives four complex roots counting multiplicity for a quartic.



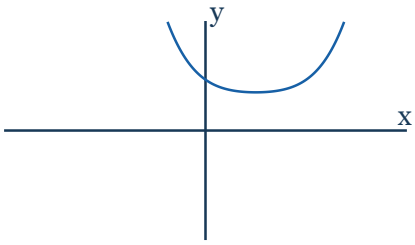
Comment on the roots.

Two real roots and two non-real conjugate roots.



Comment on the roots.

Four real roots counting multiplicity; one is repeated.



Comment on the roots.

No real roots; two pairs of non-real conjugate roots.

Example Question

Solution 29

Given $3+i$ is a root of

$2z^4 - 3z^3 - 39z^2 + 120z - 50 = 0,$

solve completely.

1. Conjugates

The conjugate $3-i$ is also a root, so

$(z-(3+i))(z-(3-i)) = z^2 - 6z + 10$

is a factor.

You could also use algebraic long division to obtain the other factor.

2. Factor

The other factor starts with $2z^2$ and ends with -5. Write $2z^2 + az - 5$.

Compare the coefficient of z^3: $a - 12 = -3$, so $a = 9$. Expanding verifies all coefficients:

$(z^2 - 6z + 10)(2z^2 + 9z - 5) = 0.$

3. Solve

$2z^2 + 9z - 5 = (2z-1)(z+5).$

The four roots are $3+i, 3-i, 1/2, -5.$

Example Question

Solution 30

Show $z^2 + 4$ is a factor of

$z^4 - 2z^3 + 21z^2 - 8z + 68,$

then solve the corresponding equation.

1. Factor

Try $(z^2+4)(z^2+az+17)$. Comparing z^3 gives $a = -2$. Expansion confirms:

$(z^2+4)(z^2-2z+17) = z^4 - 2z^3 + 21z^2 - 8z + 68.$

2. Roots

$z^2 + 4 = 0 \Rightarrow z = \pm 2i.$

For $z^2 - 2z + 17 = 0$: $(z-1)^2 = -16$, giving $z = 1 \pm 4i.$

Test Your Understanding

Solution 31

FP1 January 2010 Q6.

Given 2 and $5 + 2i$ are roots of $x^3 - 12x^2 + cx + d = 0$, $c, d \in \mathbb{R}$:

a. State the other non-real root.

b. Find c, d.

a $5 - 2i$.

b

$(x - (5 + 2i))(x - (5 - 2i)) = x^2 - 10x + 29$.

$(x - 2)(x^2 - 10x + 29) = x^3 - 12x^2 + 49x - 58$.

Hence $c = 49, d = -58$.

Exercise 1F Pearson Core Pure 1, p. 13-14

Solution 32

1. $f(z) = z^3 - 6z^2 + 21z - 26$.

a. Show $f(2) = 0$.

b. Solve $f(z) = 0$ completely.

a $f(2) = 8 - 24 + 42 - 26 = 0$.

b

Divide by $z - 2$:

$f(z) = (z - 2)(z^2 - 4z + 13)$.

The second factor gives $(z - 2)^2 = -9$.

Roots: $2, 2 + 3i, 2 - 3i$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 33

2. $f(z) = 2z^3 + 5z^2 + 9z - 6$.

a. Show $f(1/2) = 0$.

b. Write $f(z) = (2z - 1)(z^2 + bz + c)$, finding real b, c.

c. Solve $f(z) = 0$ using algebra.

b

Expand: $2z^3 + (2b - 1)z^2 + (2c - b)z - c$.

Comparing coefficients gives $2b - 1 = 5$, $-c = -6$, hence $b = 3, c = 6$; also $2c - b = 9$.

$f(z) = (2z - 1)(z^2 + 3z + 6)$.

a

$f(1/2) = 1/4 + 5/4 + 9/2 - 6 = 0$.

c

$2z - 1 = 0 \Rightarrow z = 1/2$.

$z^2 + 3z + 6 = 0 \Rightarrow z = \dfrac{-3 \pm \sqrt{-15}}{2} = -\dfrac{3}{2} \pm \dfrac{\sqrt{15}}{2} i$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 34

3. $g(z) = 2z^3 - 4z^2 - 5z - 3$. Given $z = 3$ is a root, solve $g(z) = 0$ completely.

1. Division

Division by $z - 3$:

$g(z) - 2z^2(z - 3) = 2z^2 - 5z - 3$,

$(2z^2 - 5z - 3) - 2z(z - 3) = z - 3$.

The last quotient term is 1 and the remainder is zero.

$g(z) = (z - 3)(2z^2 + 2z + 1)$.

2. Solve

$2z^2 + 2z + 1 = 0 \Rightarrow z = \dfrac{-2 \pm \sqrt{4 - 8}}{4}$

$= \dfrac{-2 \pm 2i}{4} = -\dfrac{1}{2} \pm \dfrac{1}{2} i$.

All three roots: $3, -1/2 + i/2, -1/2 - i/2$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 35

4. $p(z) = z^3 + 4z^2 - 15z - 68$. Given $-4 + i$ is a root:

a. Show $z^2 + 8z + 17$ is a factor.

b. Solve $p(z) = 0$.

a

$-4 - i$ is also a root. Their factors give

$(z + 4 - i)(z + 4 + i) = (z + 4)^2 + 1 = z^2 + 8z + 17.$

b

$p(z) = (z^2 + 8z + 17)(z - 4).$

Roots: $4, -4 + i, -4 - i.$

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 36

5. $f(z) = z^3 + 9z^2 + 33z + 25$,
$f(z) = (z + 1)(z^2 + az + b)$.

a. Find real a, b.

b. Find all three roots.

c. Find their sum.

a

Expanding gives $z^3 + (a + 1)z^2 + (a + b)z + b.$

Thus $a + 1 = 9$, $b = 25$, so $a = 8, b = 25$, and $a + b = 33.$

b

$z = -1$ or $z^2 + 8z + 25 = 0.$

Completing the square, $(z + 4)^2 = -9$, giving $z = -4 \pm 3i.$

c

$-1 + (-4 + 3i) + (-4 - 3i) = -9.$

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 37

6. $g(z) = z^3 - 12z^2 + cz + d$, $c, d \in \mathbb{R}$, has roots 6 and $3 + i$.

a. State the other non-real root.

b. Find c, d.

6a $3 - i$.

6b

$(z - 6)((z - 3)^2 + 1) = (z - 6)(z^2 - 6z + 10)$

$= z^3 - 12z^2 + 46z - 60$.

Thus $c = 46, d = -60$.

7. $h(z) = 2z^3 + 3z^2 + 3z + 1$. Given $2z + 1$ is a factor, find all roots.

7

Write $(2z + 1)(az^2 + bz + c)$. Comparing coefficients: $2a = 2, a + 2b = 3, c = 1$, so $a = b = c = 1$.

$h(z) = (2z + 1)(z^2 + z + 1)$.

Either $z = -1/2$, or

$(z + 1/2)^2 = -3/4 \Rightarrow z = -1/2 \pm (\sqrt{3}/2)i$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 38

8. $f(z) = z^3 - 6z^2 + 28z + k$, with $f(2) = 0$.

a. Find k.

b. Find the other roots.

a $8 - 24 + 56 + k = 0 \Rightarrow k = -40$.

b

$f(z) = (z - 2)(z^2 - 4z + 20)$.

The other roots satisfy $(z - 2)^2 = -16$, so $z = 2 \pm 4i$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 39

9. Find all four roots of $z^4 - 16 = 0$.

9

$z^4 - 16 = (z^2 - 4)(z^2 + 4) = (z - 2)(z + 2)(z^2 + 4)$.

Hence $z = 2, -2, 2i, -2i$.

10. $f(z) = z^4 - 12z^3 + 31z^2 + 108z - 360$.

a. Write $f(z) = (z^2 - 9)(z^2 + bz + c)$.

b. Hence find all roots.

10a

Expand: $z^4 + bz^3 + (c - 9)z^2 - 9bz - 9c$.

Thus $b = -12, c = 40$, giving

$(z^2 - 9)(z^2 - 12z + 40)$.

10b

$z = \pm 3$, or $(z - 6)^2 = -4$, so $z = 6 \pm 2i$.

Exercise 1F Pearson Core Pure 1, p. 13-14 — Solution 40

11. $g(z) = z^4 + 2z^3 - z^2 + 38z + 130$. Given $g(2 + 3i) = 0$, find all roots.

1. Conjugate factor

$2 - 3i$ is also a root, so

$(z - (2 + 3i))(z - (2 - 3i)) = z^2 - 4z + 13$

is a factor.

2. Divide

Division steps:

$g(z) - z^2(z^2 - 4z + 13) = 6z^3 - 14z^2 + 38z + 130$.

Subtract $6z(z^2 - 4z + 13)$ to leave

$10z^2 - 40z + 130$, which is $10(z^2 - 4z + 13)$.

Hence the quotient is $z^2 + 6z + 10$.

3. Roots

$g(z) = (z^2 - 4z + 13)(z^2 + 6z + 10)$.

The first factor gives $2 \pm 3i$. For the second,

$(z + 3)^2 = -1 \Rightarrow z = -3 \pm i$.

All four roots are $2 + 3i, 2 - 3i, -3 + i, -3 - i$.

12. $f(z) = z^4 - 10z^3 + 71z^2 + Qz + 442$, $Q \in \mathbb{R}$, has root $2 - 3i$.

a. Show $z^2 - 6z + 34$ is a factor.

b. Find Q.

c. Solve $f(z) = 0$.

a

$2 + 3i$ is also a root, so $z^2 - 4z + 13$ is a factor. Write

$f(z) = (z^2 - 4z + 13)(z^2 + bz + c)$.

Constant terms: $13c = 442 \Rightarrow c = 34$. Cubic terms: $b - 4 = -10 \Rightarrow b = -6$. The quadratic coefficient is $34 + 24 + 13 = 71$, as required. Thus the other factor is $z^2 - 6z + 34$.

b The linear coefficient is $-4(34) + 13(-6) = -214$, so $Q = -214$.

c

$f(z) = (z^2 - 4z + 13)(z^2 - 6z + 34)$.

The first factor gives $2 \pm 3i$. The second gives $(z - 3)^2 = -25$, hence $3 \pm 5i$.